

ARSPI-Net: An Event-Driven Reservoir-Graph Substrate for Affective EEG Perception in Simulated Embodied Control

Anonymous Author(s)

Submitted for double-blind review

Abstract—We present ARSPI-Net (Affective Reservoir-Spike Processing and Inference Network), a brain-inspired event-driven reservoir-graph substrate that transforms noisy affective electroencephalography (EEG) into a set of operationally distinct neural evidence streams and evaluates those streams as perceptual channels for a simulated embodied agent. The substrate is anchored in concrete neural mechanisms: a fixed leaky integrate-and-fire (LIF) spiking reservoir, a binned spike-count (BSC_6) event-driven code, an electrode-level temporal phase-locking (tPLV) graph, and a structure-function coupling readout κ . These produce four evidence streams: a spike-coded embedding E , dynamical descriptors D , graph-topological descriptors T , and a coupling block C . On 633 trial-averaged subject-condition observations from 211 adults under subject-grouped cross-validation across five seeds, a mechanism ablation with negative controls separates the streams: balanced accuracy is highest for the embedding-containing configurations ($E+D+T$, 0.481) and the band-power baseline (0.485), the standalone D , T , and C streams are lower but above the three-class chance level, and a shuffled-label control collapses to chance; an independent permutation-FDR protocol corroborates the ordering. The streams remain linearly distinct. We characterise the streams under four perturbation families (temporal jitter, amplitude noise, channel dropout, and graph edge perturbation) and report their operating regimes rather than a single endpoint. Finally, an explicitly defined expected-free-energy controller uses the substrate as a perceptual estimator in a simulated closed-loop affective-control task with action-determined transition noise; against passive, random, pragmatic-only, epistemic-only, and perfect-perception oracle policies, the controller reaches and confirms a target affective state, with the epistemic term adding limited marginal utility relative to pragmatic control under action-determined transitions. Clinical labels serve only as exploratory contextual validation, bounded by a false-discovery-rate result. All results are specific to the measured ERP regime.

Index Terms—EEG, neuromorphic computing, spiking neural networks, liquid state machines, reservoir computing, neural coding, predictive coding, sequential evidence accumulation, embodied AI, perception-action loops, structure-function coupling, layer ablation.

I. INTRODUCTION

EEG offers millisecond-scale access to neural activity, but scalp recordings are noisy, indirect, and subject-dependent. For an embodied agent that must perceive and act on affective state, the central engineering problem is the extraction of usable neural evidence from this partially observed, nonstationary signal. We treat the affective event-related potential (ERP) as the output of a biological dynamical system and ask how an event-driven, brain-inspired substrate can transform it

into evidence streams whose reliability can be measured under perturbation and used to drive perception under uncertainty. This is a neuromorphic signal-processing question rather than a question of psychology or diagnosis. A single endpoint classifier leaves open which part of a staged substrate carries the signal, how that signal degrades under realistic measurement perturbations, and whether two intermediate representations measure the same response under different names.

This work does not propose ARSPI-Net as a superior static affective-EEG classifier. A conventional ERP-amplitude baseline is retained as a reference and achieves higher three-class balanced accuracy under a static readout. ARSPI-Net is evaluated instead as a reservoir-graph measurement substrate: it decomposes the response into spike-coded, dynamical, graph-topological, and structure-function evidence streams; measures their perturbation-dependent operating regimes; and tests their utility as perceptual channels in a simulated embodied-control loop.

Compact convolutional baselines such as EEGNet [6], EEG graph networks [8], [10], spiking-graph hybrids [9], and multimodal attention models [11], [12] are evaluated through cross-subject protocols, ablations, and module-level metrics rather than one accuracy number. ARSPI-Net (Affective Reservoir-Spike Processing and Inference Network), a staged neuromorphic architecture for clinical affective EEG [36], [37], is evaluated here under that standard.

ARSPI-Net produces four feature families per observation: a reservoir spike embedding E , dynamical descriptors D , electrode-level topological descriptors T , and structure-function coupling descriptors C . The present study asks a structural question about these four feature families. Do E , D , T , and C behave as different measurements, or are they redundant relabellings of the same response?

Operational distinctness is one of the evaluation criteria we apply to the substrate, alongside perturbation robustness and embodied-control utility. It is a measurement-oriented test of whether the streams are empirically non-equivalent under a fixed protocol, differing in predictive sufficiency, incremental utility, label-specific sensitivity, or representational redundancy. It characterises the streams as different signal-processing views of one EEG response.

The dataset comprises a 211-subject affective EEG cohort with five overlapping psychiatric labels and per-subject comorbidity counts. The architecture decomposes the staged pipeline into four blocks, each tied to a named neural mechanism

rather than treated as a black-box feature extractor. The evaluation protocol combines subject-respecting permutation inference, FDR correction, redundancy under centered-kernel alignment [17], an electrode-permutation null on the coupling readout, and a sequential-decision simulation that places the four blocks in an embodied perception loop. The simulation is a precursor to active inference [29]–[31]: closed-loop on the belief side, open-loop on the environment side.

This paper contributes four technical elements. First, it specifies an event-driven reservoir-graph substrate in which each evidence stream is anchored to a named neural mechanism (LIF spiking, BSC₆ event-driven coding, tPLV connectivity, and κ structure-function coupling), with an explicit mathematical observable for each. Second, it establishes the operational distinctness of the streams through a mechanism ablation (A0 to A9) with negative controls, distinguishing predictive utility (whether a stream raises endpoint accuracy) from measurement utility (whether a stream exposes structure the others omit). Third, it quantifies the perturbation robustness of the streams under temporal jitter, amplitude noise, channel dropout, and graph edge perturbation, reporting operating regimes rather than a single endpoint. Fourth, it evaluates the streams as perceptual channels for a simulated embodied affective-control loop driven by an explicitly defined expected-free-energy controller, compared against passive, random, pragmatic-only, epistemic-only, and oracle baselines. Throughout, dataset provenance and quality control are reported, clinical labels are used only as exploratory validation bounded by a false-discovery-rate result, and results are reported as aggregate, deidentified outputs under data-governance constraints.

II. RELATED WORK

Brain-inspired computing treats biophysical neuron and synapse models as a computational substrate rather than an analogy. The leaky integrate-and-fire neuron is an idealised reduction of the Hodgkin-Huxley membrane [4]; liquid state machines [1] and echo state networks [2] use a fixed recurrent reservoir as a high-dimensional projection for simple readouts [3], and event-driven substrates such as Loihi [34] and SpiNNaker [35] run these primitives natively. Graph neural networks model EEG as electrode graphs but diverge in node features, graph construction, and readout [8], and spiking-graph hybrids add event-based temporal computation [9]. ARSPI-Net departs from end-to-end EEG classifiers by treating the reservoir as a fixed nonlinear measurement operator whose outputs feed several downstream descriptors rather than one classification head.

This paper adopts the standard EEG-evaluation practice of cross-subject protocols, controlled ablation, and claim-bounded clinical discussion [6], [7], [9], [11], and additionally evaluates the architecture as competing perceptual policies for a simulated embodied agent, closer to active-inference accounts of perception-action [29]–[31]. The present study evaluates whether the four feature families behave as operationally distinct measurement layers and as perceptual channels under perturbation and embodied control.

III. ARSPI-NET LAYER FORMULATION

A. Input Representation

Let

$$X_s^{(c)} \in \mathbb{R}^{N_{ch} \times T}, \quad N_{ch} = 34, \quad (1)$$

denote the preprocessed event-related EEG response for subject s under condition $c \in \{\text{negative, neutral, pleasant}\}$. The affective task uses subject-condition observations directly. Clinical-label analyses average each feature family over the available conditions per subject.

B. Reservoir Dynamics and Spike Embedding

Each channel is fed independently into a fixed LIF reservoir whose discrete-time update is

$$\mathbf{u}_{t+1} = \beta \mathbf{u}_t + W_{in} x_t + W_{rec} \mathbf{z}_t - \mathbf{r}_t, \quad (2)$$

with spike emission

$$\mathbf{z}_{t+1} = H(\mathbf{u}_{t+1} - \theta), \quad (3)$$

where $H(\cdot)$ is the Heaviside step function, $\beta = 0.05$ is the membrane decay, $\theta = 0.5$ is the threshold, W_{in} and W_{rec} are Xavier-initialised input and recurrent matrices, and the reservoir size is $N_{res} = 256$. The recurrent matrix is rescaled so that its largest eigenvalue magnitude is 0.9. Spike counts are pooled into $n_{bins} = 6$ non-overlapping windows over the response interval $t \in [10, 70]$ to form the BSC₆ feature, and the per-channel BSC₆ vector is PCA-compressed to 64 components. The full embedding for a subject-condition observation is

$$E_s^{(c)} = [h_1 \| h_2 \| \dots \| h_{34}] \in \mathbb{R}^{2176}, \quad (4)$$

where $h_v \in \mathbb{R}^{64}$ is the channel-level embedding.

C. Dynamical, Topological, and Coupling Blocks

The dynamical block D contains reservoir-response descriptors per channel. The topological block T summarises electrode-level connectivity as a temporal phase-locking matrix and reduces it to two per-electrode summaries (mean strength and clustering). For per-channel matrices $D_s \in \mathbb{R}^{34 \times p}$ and $T_s \in \mathbb{R}^{34 \times q}$, the rank-correlation coupling matrix is

$$K_s = \text{corr}_\rho(D_s, T_s) \in \mathbb{R}^{p \times q} \quad (5)$$

and the coupling magnitude is

$$\kappa_s = \frac{\|K_s\|_F}{\sqrt{pq}}. \quad (6)$$

The block C contains κ_s alongside signed strength and clustering coupling summaries.

D. Operational Distinctness Criteria

For families $L_i, L_j \in \{E, D, T, C\}$ and target Y , the performance contrast is

$$\Delta_{ij}(Y) = \mathcal{M}(f(L_i), Y) - \mathcal{M}(f(L_j), Y), \quad (7)$$

with $\mathcal{M}$ either balanced accuracy or ROC-AUC and f a fixed L2-regularised logistic readout. The incremental utility relative to a baseline L_b is

$$\Gamma_i(Y|L_b) = \mathcal{M}(f([L_b|L_i]), Y) - \mathcal{M}(f(L_b), Y). \quad (8)$$

Redundancy is assessed using linear centered-kernel alignment,

$$\text{CKA}(X, Y) = \frac{\|X_c^\top Y_c\|_F^2}{\|X_c^\top X_c\|_F \|Y_c^\top Y_c\|_F}, \quad (9)$$

with X_c and Y_c column-centered [17].

IV. BRAIN-INSPIRED MECHANISMS AND EMBODIED INTERPRETATION

The blocks of ARSPI-Net each have a named neural correlate. This section states the correlate, the implementation choice, and its scope, so that the architecture is described as a sequence of physiologically motivated transformations rather than a stack of feature extractors. Table I summarises the mapping from each neural principle to its ARSPI-Net implementation, the computational object it produces, the observable it exposes, and its relevance to an embodied agent.

A. LIF Spiking Reservoir as a Recurrent Cortical Substrate

The reservoir realised by (2)–(3) is a fixed leaky integrate-and-fire network. The neuron model is the canonical biophysical reduction of the Hodgkin-Huxley axon [4]: a sub-threshold membrane that integrates synaptic drive, decays toward rest, fires when the integrated potential crosses threshold, and resets. With $\beta = 0.05$ the decay constant places the reservoir in the temporal regime that liquid state machines use for transient projection of input streams [1]. The recurrent matrix is held at spectral radius 0.9, just below the marginal-stability boundary at which a recurrent network’s transient response becomes long enough to support readout while remaining contractive on average [2], [5]. The reservoir weights are fixed; supervised training is confined to the downstream readout.

The fixed-reservoir design follows the reservoir-computing abstraction [3], in which the recurrent transformation is held fixed while the readout is trained. The fixed-reservoir design isolates the recurrent transformation from the supervised readout.

B. BSC₆ as an Event-Driven Code

The embedding E in (4) is built from binned spike counts over $n_{\text{bins}} = 6$ windows of the post-stimulus interval $t \in [10, 70]$. BSC₆ has three relevant properties. First, it is event-driven. The representation is read from spikes rather than membrane traces; the readout operates on spike-count summaries rather than membrane trajectories. Second, the encoding window starts a small distance after stimulus onset and ends well before the stimulus offset, covering the temporal

interval where ERP components such as P200, N200, and the early portion of P300 are expressed. The bins are coarse enough to be robust to small temporal jitter but fine enough to preserve coarse temporal structure within the response. Third, the format is compatible with event-counter abstractions used in neuromorphic systems: spike counts over fixed time windows correspond to on-chip event-counter operations on Loihi [34] and SpiNNaker [35], and binned spike-count classifiers have been shown to support unsupervised digit recognition [33]. PCA compression to 64 per channel gives a fixed-length representation that the downstream readout consumes without retraining the reservoir.

C. tPLV Connectivity as Cortical Network Organisation

The topological block T is derived from per-observation 34×34 temporal phase-locking value matrices [23]. PLV measures inter-electrode phase consistency across trials and is one of the standard tools in EEG functional connectivity. Computed within the event-related response window, tPLV captures the connectivity that is in force during the task rather than during rest, mitigating the stationarity concerns that follow from full-trace PLV estimates. The matrix is reduced to two per-electrode summaries, mean strength and clustering. Both belong to the established graph-theoretic vocabulary of brain network analysis [24], and both have a direct biological reading: strength tracks how synchronised an electrode is with the rest of the head, while clustering tracks whether its synchronised neighbours are themselves synchronised. The reduction loses information by design. We trade pairwise resolution for a per-channel descriptor compatible with the embedding shape used by the other blocks.

D. Structure-Function Coupling κ

The coupling readout (6) measures whether a subject’s per-electrode dynamical signature (D) and topological signature (T) co-vary across electrodes. It is a single non-negative scalar bounded above by 1, computed per subject-condition pair, and admits an electrode-permutation null. Structure-function coupling is a recurring quantity in network neuroscience, where it indexes the degree to which functional patterns line up with the structural backbone [25]–[27]. The use here is closer to a within-subject signature than to a population-level claim: κ asks whether one subject’s local electrode dynamics and the same subject’s electrode-level network position are coordinated, and the permutation null asks whether the coordination is more than would arise from chance electrode labelling. The block C embeds κ alongside signed strength and clustering coupling.

E. Sequential Evidence Accumulation

The four streams are interpreted as operationally distinct sub-systems available to an embodied agent (Section VI-D). The agent maintains a Dirichlet posterior $\text{Dir}(\alpha_t)$ over a held-out subject’s affective response; at each step it picks one of the four feature-stream policies and updates the posterior under the soft observation

$$\alpha_{t+1} = \alpha_t + \mathbf{p}_{\theta_a}(x_t), \quad (10)$$

TABLE I
NEURAL MECHANISMS IMPLEMENTED BY ARSPI-NET, EACH ANCHORED TO A CONCRETE COMPUTATIONAL OBJECT AND A MEASURED OBSERVABLE.

Neural principle	Implementation	Computational object	Measured observable	Relevance to embodied AI
Event-driven coding	BSC ₆ binning	temporal-bin count vector	bin evidence	sparse perceptual observation
Recurrent cortical dynamics	LIF reservoir	reservoir state trajectory	embedding E	nonlinear temporal transformation
Temporal state statistics	descriptor block D	trajectory descriptors	variance, autocorrelation, complexity	dynamical evidence stream
Functional connectivity	tPLV graph	adjacency / graph operator	topology descriptors T	spatial dependency
Structure-function coupling	κ	coupling readout	shuffle-null significance	systems-level coordination
Embodied perception	posterior update + policy	belief state	entropy, success, steps	closed-loop perceptual utility

where $\mathbf{p}_{\theta_a}(x_t)$ is the policy- a classifier’s class-probability vector for x_t , and expected information gain is the KL divergence of the predictive distribution across the update. This is a sequential-decision counterpart to the offline ablation: which stream most rapidly reduces an agent’s posterior uncertainty, closed-loop on the belief side and open-loop on the environment side, a precursor to active inference [29], [30].

V. EXPERIMENTAL PROTOCOL

A. Data, Splitting, and Privacy

The analysis uses 633 subject-condition observations from 211 community-recruited adults, exactly balanced across the three affective conditions (negative, neutral, pleasant). Clinical-label analyses use 206 subjects with available SUD, MDD, PTSD, GAD, and ADHD labels. Comorbidity is the rule rather than the exception in this cohort: most subjects carry more than one label, and the analyses below use comorbidity counts as a covariate where appropriate. Subject-level data are restricted; reported outputs are aggregate and deidentified. Raw EEG, clinical metadata, and subject-level features remain access-controlled.

Table II summarises the analysis protocol.

B. Architecture Overview

Fig. 1 shows the staged ARSPI-Net architecture.

C. Feature Blocks and Diagnostics

Table III lists the evaluated blocks. The embedding E has shape (633, 2176) with full standardised rank 632 and no constant columns.

D. Ablation Matrices

Affective classification used the A0 to A9 configurations. The classifier is L2-regularised logistic regression with train-fold-only standardisation under a 10-fold StratifiedGroupK-Fold subject separation. Clinical-label sensitivity used the C1 to C6 configurations at the subject level with the same logistic readout. Linear separability is the deliberate choice, because operational distinctness is a property of the feature spaces

under a fixed read-out, not of an unconstrained nonlinear endpoint model. The optimisation is

$$\hat{\theta} = \arg \min_{\theta} \left[-\sum_{i=1}^n \log p_{\theta}(y_i|x_i) + \lambda \|\theta\|_2^2 \right] \quad (11)$$

and the primary metric is balanced accuracy

$$\text{BA} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FN_k} \quad (12)$$

with $K = 2$ for clinical-label sensitivity and $K = 3$ for affective classification.

VI. RESULTS

We organise the evidence as one argument over four pillars: (A) dataset provenance and the computational objects the substrate produces; (B) the reservoir-graph mechanism ablation; (C) perturbation robustness and operating regimes; and (D) the simulated embodied affective-control loop. The primary ablation is the multi-seed mechanism ablation of Section VI-B; an independent ten-fold permutation-FDR protocol corroborates it, and that protocol together with the secondary clinical, redundancy, and comorbidity analyses is reported in the supplementary material.

A. Dataset Provenance and Observed Computational Objects

The data are trial-averaged event-related EEG from 211 community-recruited adults (subject 127 excluded for a recording anomaly), giving 633 subject-condition observations balanced across Negative, Neutral, and Pleasant. The evaluation uses subject-grouped cross-validation, so each subject appears in either the training or the test partition of a fold. Subject-level data are restricted; reported outputs are aggregate and deidentified.

Fig. 2 shows the computational objects the substrate produces, for one exemplar per affective class: the LIF spike raster and membrane trajectory that generate the embedding E (panels a and b), the per-class 34×34 tPLV connectivity that defines the graph block T (panel c), and the BSC₆ projection that defines the event-driven code (panel d). Structure-function coupling is summarised by $\kappa = \|C\|_F / \sqrt{pq}$: across the 633

TABLE II
AFFECTIVE AND CLINICAL ANALYSIS PROTOCOL

Stage	Input	Operation	Output / Control
Data	211 subjects, 633 subject-condition	Condition-level affective analysis;	Three affective classes; 206 subjects
Feature diagnostics	EEG observations	subject-level clinical analysis	with clinical labels
Affective ablation	BandPower, E , D , T , C	Check finite values, ranks, zero	Embedding E nonzero with standard-
		fractions, constant columns	ised rank 632; all blocks nondegenerate
	A0 to A9 feature configurations	10-fold subject-grouped CV with	Balanced accuracy, macro ROC-AUC,
		train-fold-only scaling	macro-F1, predictions, confusion matrices
Clinical sensitivity	C1 to C6 subject-average features	Binary logistic readout per clinical	Exploratory clinical-label sensitivity,
		label	not diagnostic validation

ARSPI-Net Pipeline: From Raw EEG to Graph-Based Clinical Analysis

Row 1: Multichannel EEG input (211 subjects $\times$ 3 conditions). Row 2: LIF reservoir transformation (EEG $\rightarrow$ spikes $\rightarrow$ BSC₆ $\rightarrow$ PCA-64). Row 3: Graph construction and inter-channel structure.

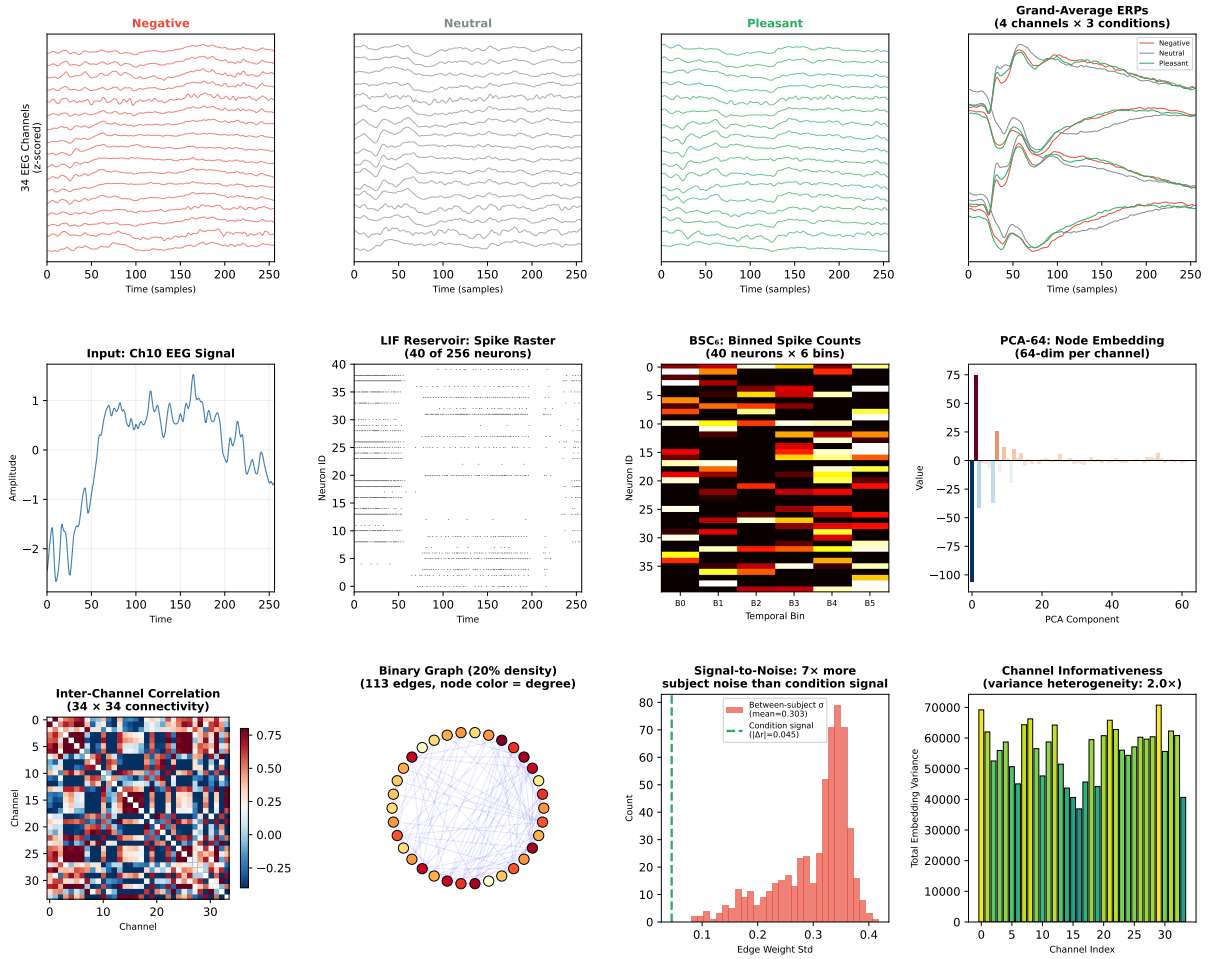


Fig. 1. ARSPI-Net pipeline overview: the staged event-driven reservoir-graph architecture, from the per-channel LIF reservoir and BSC₆ event-driven code to the embedding E , dynamical descriptors D , topological descriptors T , and coupling readout C evaluated in this work.

observations the mean is 0.250 (sd 0.115), and a 5,000-permutation electrode-shuffle null is rejected at $p < 0.05$ in 33.5% of observations, so a non-trivial subset of subjects show coordination above a randomised electrode labelling. Subject-mean κ is uncorrelated with per-subject classification accuracy ($\rho = 0.01$), behaving as a complementary systems-

level readout rather than a proxy for predictive sufficiency.

B. Reservoir-Graph Mechanism Ablation

The primary ablation evaluates configurations A0 to A9 under subject-grouped cross-validation across five seeds, with bootstrap intervals and two negative controls (Table IV, Fig. 3).

TABLE III
FEATURE FAMILIES USED IN THE OPERATIONAL-DISTINCTNESS ANALYSIS

Block	Dimension	Source	Operational role
BandPower	170	Spectral features	Conventional frequency-domain comparator
E	2176	BSC ₆ /PCA-64 reservoir embedding	Temporal spike representation
D	238	Reservoir trajectory descriptors	Dynamical response characterization
T	68	Graph-topological descriptors	Spatial organization across electrodes
C	3	Structure-function coupling	Alignment of dynamics and topology

Brain-inspired reservoir dynamics underlying ARSPI-Net (LIF: $N_{res} = 256$, $\beta = 0.05$, $\theta = 0.5$; BSC6 window $t \in [10, 70]$)

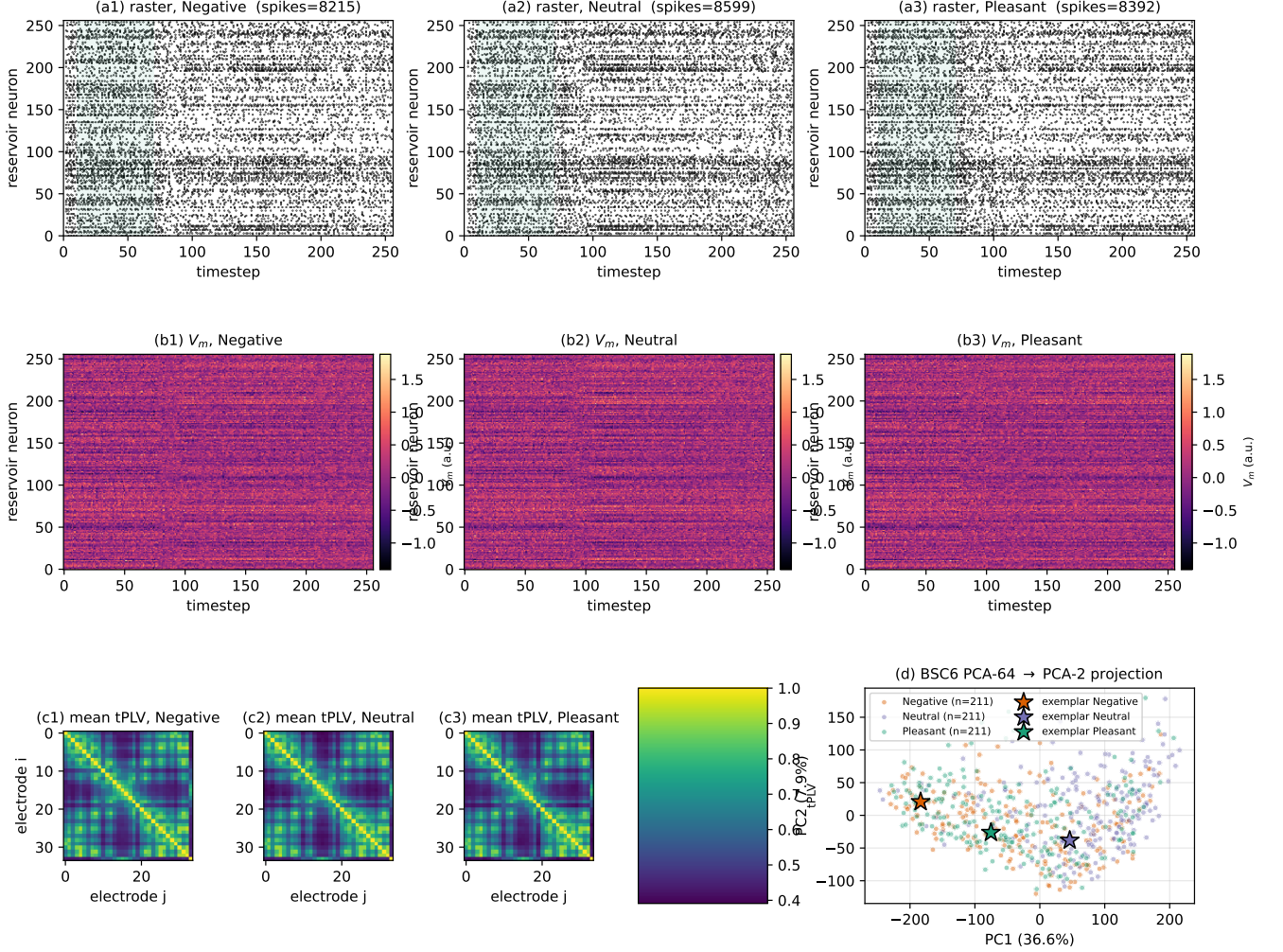


Fig. 2. Computational objects produced by the substrate. (a) LIF spike rasters for one exemplar per affective class on a representative channel; the BSC₆ encoding window $t \in [10, 70]$ is shaded. (b) Membrane-potential heatmaps for the same exemplars. (c) Per-class average 34×34 tPLV connectivity matrices across all 211 subjects. (d) BSC₆ PCA projection (PC1 vs PC2) coloured by class with the three exemplars marked.

Balanced accuracy is highest for the spectral band-power baseline (A0, 0.485) and the embedding-containing configurations ($E+D+T$ 0.481, $E+D+T+C$ 0.481, $E+D$ 0.478, E 0.463); the standalone dynamical, topological, and coupling streams are lower (D 0.432, C 0.368, T 0.355) but remain above the three-class chance level of 0.333. The shuffled-label control collapses to chance (0.335), confirming the protocol. An independent ten-fold permutation-FDR protocol corroborates the ordering: eight of ten configurations exceed a subject-

respecting permutation null, with standalone T and C the two exceptions.

The ablation shows operational differentiation rather than classifier superiority. The embedding carries most of the affective-classification signal, while D , T , and C remain linearly distinct (largest pairwise centred-kernel alignment $\text{CKA}(E, D) = 0.31$) and expose dynamical, topological, and coupling structure that the embedding alone omits. Clinical labels are used only as exploratory contextual validation: per-

TABLE IV

MECHANISM ABLATION UNDER SUBJECT-GROUPED CROSS-VALIDATION. BALANCED ACCURACY (BA) AND MACRO-F1 ARE REPORTED AS MEAN WITH A BOOTSTRAP 95% INTERVAL. CHANCE BA = 0.333. EACH ROW REPORTS THE FEATURE CONFIGURATION, THE NEURAL MECHANISM IT CARRIES, AND ITS FEATURE DIMENSION.

Cfg	Mechanism	Dim	BA	95% CI	macro-F1
A0	spectral band-power (non-neuromorphic baseline)	170	0.485	[0.471, 0.500]	0.483
A1	LIF reservoir spike embedding (E)	2176	0.463	[0.450, 0.476]	0.460
A2	reservoir dynamical descriptors (D)	238	0.432	[0.413, 0.450]	0.429
A3	tPLV graph topology descriptors (T)	68	0.355	[0.343, 0.367]	0.353
A4	structure-function coupling (C)	3	0.368	[0.355, 0.382]	0.352
A5	D+T (dynamics + topology)	306	0.439	[0.425, 0.451]	0.436
A6	E+D (embedding + dynamics)	2414	0.478	[0.465, 0.490]	0.475
A7	E+T (embedding + topology)	2244	0.464	[0.453, 0.476]	0.461
A8	E+D+T (full reservoir-graph substrate)	2482	0.481	[0.468, 0.494]	0.478
A9	E+D+T+C (substrate + coupling readout)	2485	0.481	[0.468, 0.493]	0.478
CTRL_shuffled_label	negative control	2482	0.335	[0.320, 0.351]	0.333
CTRL_shuffled_subject	negative control	2482	0.474	[0.461, 0.487]	0.471

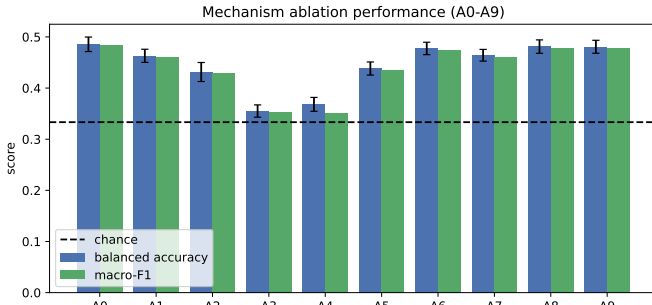


Fig. 3. Mechanism-ablation performance for configurations A0 to A9. Balanced accuracy (with bootstrap intervals) and macro-F1 are shown against the three-class chance level. Bars summarise the measured cross-validated performance of each evidence stream and combination.

diagnosis balanced accuracies are near chance (0.49 to 0.53 across SUD, MDD, PTSD, GAD, ADHD) and clinical contrasts are bounded by a Benjamini-Hochberg false-discovery-rate result across the 30 diagnosis $\times$ configuration cells, serving as exploratory context only. The clinical-sensitivity, redundancy, and comorbidity-adjusted analyses are reported in full in the supplementary material.

C. Perturbation Robustness and Operating Regimes

We characterise each evidence stream under four perturbation families (temporal jitter, amplitude noise, channel dropout, and graph edge perturbation) applied at the representation level across all ten configurations, with a limited raw-signal diagnostic on a prespecified subset that recomputes a reservoir embedding within fold. Fig. 4 reports the degradation profiles, which separate the streams by operating regime rather than by a single endpoint. Under additive amplitude noise the spectral band-power baseline is the most resilient, retaining 0.395 balanced accuracy at 5 dB SNR, whereas the $E+D+T$ configuration falls toward chance (0.341, from a clean 0.481); the same ordering holds under channel dropout (0.408 versus 0.340 at 30% dropout). Graph edge perturbation, by contrast, leaves the embedding-dominated configurations essentially unchanged ($E+D+T$: 0.481 $\rightarrow$ 0.483 at 30%) while degrading the topological stream (T : 0.355 $\rightarrow$ 0.331).

The measured regime is stream-specific: band-power and embedding descriptors are comparatively robust to signal-domain corruption, and the graph descriptor is sensitive to graph-domain perturbation. A supplemental nested evidence-routing analysis further bounds this result: an oracle over the streams shows large separable headroom, but label-free stream selection remains at or below the best fixed stream under the measured perturbation regime, supporting the interpretation of the substrate as making the operating regimes observable.

A classical ERP-amplitude baseline (windowed $P1$ to LPP features) reaches 67.2% balanced accuracy on the three-class task, exceeding ARSPI-Net under a static readout. This comparison anchors the endpoint-classification trade-off: ARSPI-Net is not optimized as a static classifier, but as a reservoir-graph measurement substrate that exposes operationally distinct, perturbation-characterised evidence streams with simulated embodied-control utility. The full static comparison is reported in the supplementary material.

D. Simulated Embodied Affective-Control

We evaluate the substrate as the perceptual estimator for a simulated embodied agent. The agent maintains a categorical belief b over the affective state, updated by sequential Bayesian accumulation of the per-policy classifier likelihood (Section IV); the environment is action-determined with transition noise, $P(s' | a) = (1 - \epsilon) \mathbf{1}[s'=a] + \epsilon \mathcal{U}$. The loop is evaluated as a simulation over recorded observations.

Open-loop, the four streams act as competing observation channels: under sequential evidence accumulation their final posterior entropy ranks them $E+D+T$ (fastest, 0.931 nats) $< E < D < \text{random} < T$ (slowest, 1.075), the same ordering as the offline ablation, so offline sufficiency and perceptual-channel value agree.

Closed-loop, the agent must drive the environment to a target affective state and confirm it. An expected-free-energy (EFE) controller selects the action $a^* = \arg \min_a \text{EFE}(a)$ minimising

$$\text{EFE}(a) = \underbrace{\text{KL}(P(s' | a) \| C)}_{\text{risk}} + \underbrace{\mathbb{E}_{s' \sim P(\cdot | a)}[\hat{H}(s')]}_{\text{ambiguity}}, \quad (13)$$

Robustness degradation by perturbation (representation level)

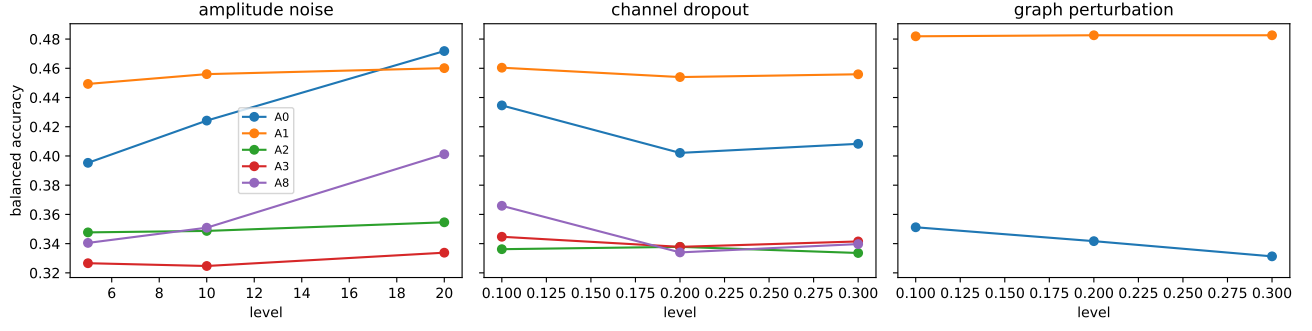


Fig. 4. Representation-level robustness degradation under amplitude noise, channel dropout, and graph edge perturbation for representative configurations. Curves report balanced accuracy as a function of perturbation level; they expose the operating regime of each evidence stream rather than a single endpoint.

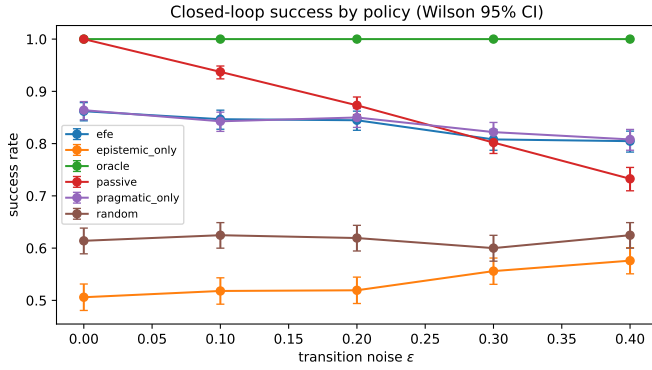


Fig. 5. Closed-loop success rate by policy and transition noise ϵ , with Wilson 95% intervals. The expected-free-energy and pragmatic controllers reach and confirm the target affective state and maintain higher success rates across transition-noise levels; belief-blind, random, and epistemic-only baselines remain below the EFE and pragmatic controllers under the measured regime.

where C is a preference distribution peaked on the target and $\hat{H}(s)$ is the expected observation-posterior entropy for state s estimated on the training fold; the agent stops when $b(\text{target}) \geq 0.8$. It is compared against passive (belief-blind), random, pragmatic-only (risk term only), epistemic-only (ambiguity term only), and a perfect-perception oracle, over 1,500 episodes per policy at each transition-noise level ϵ (Table V, Fig. 5). The oracle bounds success at 1.000. The EFE and pragmatic controllers reach and confirm the target (0.862 and 0.864 at $\epsilon = 0$; 0.805 and 0.808 at $\epsilon = 0.4$), while belief-blind control degrades fastest with transition noise (1.000 $\rightarrow$ 0.733) and the random and epistemic-only baselines stay low. Under an action-determined transition the epistemic term adds limited marginal utility, so the EFE controller matches pragmatic-only control, and the gap to the oracle quantifies the cost of single-trial perceptual unreliability.

VII. DISCUSSION

The mechanism ablation establishes operational differentiation. Embedding-containing configurations carry most of the affective-classification signal, the standalone dynamical, topological, and coupling streams are weaker classifiers yet

TABLE V

CLOSED-LOOP POLICY COMPARISON. SUCCESS RATE (WILSON 95% CI), MEAN STEPS, AND FINAL POSTERIOR ENTROPY BY TRANSITION NOISE ϵ .

Policy	ϵ	Success	Mean steps	Final $\mathcal{H}$
EFE	0.0	0.862 [0.844, 0.878]	5.00	0.210
EFE	0.1	0.847 [0.828, 0.864]	5.42	0.211
EFE	0.2	0.845 [0.826, 0.862]	5.54	0.220
EFE	0.3	0.808 [0.787, 0.827]	5.94	0.209
EFE	0.4	0.805 [0.784, 0.824]	6.12	0.197
Epistemic-only	0.0	0.506 [0.481, 0.531]	8.86	0.122
Epistemic-only	0.1	0.518 [0.493, 0.543]	8.78	0.122
Epistemic-only	0.2	0.519 [0.494, 0.544]	8.88	0.141
Epistemic-only	0.3	0.556 [0.531, 0.581]	8.54	0.139
Epistemic-only	0.4	0.576 [0.551, 0.601]	8.44	0.152
Oracle	0.0	1.000 [0.997, 1.000]	1.00	0.000
Oracle	0.1	1.000 [0.997, 1.000]	1.07	0.000
Oracle	0.2	1.000 [0.997, 1.000]	1.16	0.000
Oracle	0.3	1.000 [0.997, 1.000]	1.23	0.000
Oracle	0.4	1.000 [0.997, 1.000]	1.37	0.000
Passive	0.0	1.000 [0.997, 1.000]	1.00	0.459
Passive	0.1	0.937 [0.924, 0.949]	1.00	0.474
Passive	0.2	0.873 [0.856, 0.889]	1.00	0.478
Passive	0.3	0.802 [0.781, 0.821]	1.00	0.464
Passive	0.4	0.733 [0.710, 0.754]	1.00	0.478
Pragmatic-only	0.0	0.864 [0.846, 0.880]	4.95	0.207
Pragmatic-only	0.1	0.843 [0.823, 0.860]	5.55	0.209
Pragmatic-only	0.2	0.850 [0.831, 0.867]	5.45	0.217
Pragmatic-only	0.3	0.822 [0.802, 0.841]	5.68	0.205
Pragmatic-only	0.4	0.808 [0.787, 0.827]	6.04	0.206
Random	0.0	0.614 [0.589, 0.638]	8.17	0.186
Random	0.1	0.625 [0.600, 0.649]	8.20	0.186
Random	0.2	0.619 [0.595, 0.644]	8.25	0.186
Random	0.3	0.600 [0.575, 0.625]	8.42	0.179
Random	0.4	0.625 [0.600, 0.649]	8.10	0.184

remain linearly distinct ($\text{CKA}(E, D) = 0.31$) and expose structure the embedding omits, while a shuffled-label control collapses to chance. A stream that is sub-threshold as a classifier can remain operationally distinct as a measurement view. The brain-inspired claim rests on the named mechanisms of Section IV (LIF spiking, BSC₆ event-driven coding, tPLV connectivity, and κ coupling) rather than on analogy.

Perturbation robustness locates a stream-specific operating regime rather than a single ordering: the band-power and embedding descriptors are comparatively robust to signal-domain corruption (amplitude noise, channel dropout), while the topological descriptor is the component sensitive to graph-

domain perturbation. Resource accounting in the supplementary material reports reservoir size, event rate, spike sparsity, and runtime (the recurrent layer performs on the order of the measured spike-sparsity fraction of the dense-equivalent operation count, mappable onto a small number of event-driven neuro-cores [34], [35]); these are computational measurements of the event-driven substrate.

The simulated embodied-control result establishes that the substrate functions as a usable perceptual estimator: an expected-free-energy controller reaches and confirms a target affective state and maintains higher success rates with transition noise, bounded above by a perfect-perception oracle. Under an action-determined transition the epistemic term adds limited marginal utility, so the expected-free-energy controller matches pragmatic control, and the oracle gap quantifies the cost of single-trial perceptual unreliability. The loop is a sequential-decision precursor to active inference [29], [30], closed-loop on the belief side and open-loop on the environment side, evaluated as a simulation over recorded observations.

VIII. LIMITATIONS

The embodied evaluation is a closed-loop simulation over recorded EEG observations. The agent updates beliefs and selects policies sequentially, but the environment is not a physical robot and the EEG stream is not acquired online. The result therefore supports simulated perceptual-control utility, not deployed robotic embodiment.

All measurements are restricted to the measured ERP regime. The access-controlled dataset limits direct public benchmarking, but it provides the properties required for this study: a 211-subject affective ERP cohort, clinically annotated metadata, balanced subject-condition observations, and subject-grouped validation. Public emotion-EEG datasets such as SEED or DEAP support different benchmarking questions, but they do not replace the clinically annotated measured ERP regime evaluated here.

The clinical-label analyses are exploratory. Near-chance performance across SUD, MDD, PTSD, GAD, and ADHD is consistent with the cohort structure: comorbidity is common, diagnostic labels are overlapping, and the substrate was designed to measure affective ERP evidence streams rather than to validate diagnostic biomarkers. The false-discovery-rate result therefore bounds the clinical interpretation rather than weakening the engineering contribution.

IX. CONCLUSION

ARSPI-Net is a brain-inspired event-driven reservoir-graph substrate that transforms affective EEG into operationally distinct evidence streams: a spike-coded embedding, dynamical descriptors, a phase-locking graph, and a structure-function coupling readout. A multi-seed mechanism ablation with negative controls separates the streams; a perturbation analysis reports their operating regimes; and a simulated embodied affective-control loop with an explicitly defined expected-free-energy controller evaluates them as perceptual channels. The streams behave as operationally distinct measurement views

of one EEG response, and the substrate supports the brain-inspired-computing-for-embodied-AI framing on grounds of mechanism and of a simulated closed-loop demonstration, with clinical claims bounded and scope restricted to the measured regime.

DATA AND CODE AVAILABILITY

The study dataset is access-controlled and available from the originating clinical research laboratory subject to approval and a data-use agreement. Raw EEG, clinical metadata, and subject-level feature artifacts are not publicly distributed. Reported outputs are aggregate and deidentified. The manuscript and supplementary material provide the mathematical definitions, model configuration, evaluation protocol, and aggregate results needed to evaluate the reported analyses.

ACKNOWLEDGMENT

Acknowledgments are withheld for double-blind review.

REFERENCES

- [1] W. Maass, T. Natschl ger, and H. Markram, "Real-time computing without stable states: A new framework for neural computation based on perturbations," *Neural Computation*, vol. 14, no. 11, pp. 2531–2560, 2002.
- [2] H. Jaeger, "The echo state approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
- [3] M. Luko evi ius and H. Jaeger, "Reservoir computing approaches to recurrent neural network training," *Computer Science Review*, vol. 3, no. 3, pp. 127–149, 2009.
- [4] A. L. Hodgkin and A. F. Huxley, "A quantitative description of membrane current and its application to conduction and excitation in nerve," *Journal of Physiology*, vol. 117, no. 4, pp. 500–544, 1952.
- [5] D. Sussillo and L. F. Abbott, "Generating coherent patterns of activity from chaotic neural networks," *Neuron*, vol. 63, no. 4, pp. 544–557, 2009.
- [6] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C.-P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, Art. no. 056013, 2018.
- [7] A. Craik, Y. He, and J. L. Contreras-Vidal, "Deep learning for electroencephalogram (EEG) classification tasks: A review," *Journal of Neural Engineering*, vol. 16, no. 3, Art. no. 031001, 2019.
- [8] D. Klepl, M. Wu, and F. He, "Graph neural network-based EEG classification: A survey," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 493–503, 2024.
- [9] P. Gong, P. Wang, Y. Zhou, and D. Zhang, "A spiking neural network with adaptive graph convolution and LSTM for EEG-based brain-computer interfaces," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 1440–1450, 2023.
- [10] D. Klepl, F. He, M. Wu, D. J. Blackburn, and P. Sarrigiannis, "Adaptive gated graph convolutional network for explainable diagnosis of Alzheimer's disease using EEG data," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 31, pp. 3978–3987, 2023.
- [11] Y. Guo, K. Yang, and Y. Wu, "A multi-modality attention network for driver fatigue detection based on frontal EEG, EDA and PPG signals," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 6, pp. 4009–4022, 2025.
- [12] H.-S. Chiang, M.-Y. Chen, and L.-S. Liao, "Cognitive depression detection cyber-medical system based on EEG analysis and deep learning approaches," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 2, pp. 608–616, 2023.
- [13] F. Del Pup, A. Zanola, L. F. Tshimanga, A. Bertoldo, and M. Atzori, "The more, the better? Evaluating the role of EEG preprocessing for deep learning applications," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 33, pp. 1061–1070, 2025.
- [14] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. International Conference on Learning Representations*, 2017.

- [15] P. Veličković *et al.*, “Graph attention networks,” in *Proc. International Conference on Learning Representations*, 2018.
- [16] P. W. Battaglia *et al.*, “Relational inductive biases, deep learning, and graph networks,” arXiv:1806.01261, 2018.
- [17] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton, “Similarity of neural network representations revisited,” in *Proc. International Conference on Machine Learning*, pp. 3519–3529, 2019.
- [18] Y. Benjamini and Y. Hochberg, “Controlling the false discovery rate: A practical and powerful approach to multiple testing,” *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.
- [19] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ, USA: Lawrence Erlbaum Associates, 1988.
- [20] A. Subasi, “EEG signal classification using wavelet feature extraction and a mixture of expert model,” *Expert Systems with Applications*, vol. 32, no. 4, pp. 1084–1093, 2007.
- [21] B. Hjorth, “EEG analysis based on time domain properties,” *Electroencephalography and Clinical Neurophysiology*, vol. 29, no. 3, pp. 306–310, 1970.
- [22] S. Makeig *et al.*, “Dynamic brain sources of visual evoked responses,” *Science*, vol. 295, no. 5555, pp. 690–694, 2002.
- [23] J.-P. Lachaux, E. Rodriguez, J. Martinerie, and F. J. Varela, “Measuring phase synchrony in brain signals,” *Human Brain Mapping*, vol. 8, no. 4, pp. 194–208, 1999.
- [24] E. Bullmore and O. Sporns, “Complex brain networks: Graph theoretical analysis of structural and functional systems,” *Nature Reviews Neuroscience*, vol. 10, no. 3, pp. 186–198, 2009.
- [25] C. J. Honey, O. Sporns, L. Cammoun, X. Gigandet, J. P. Thiran, R. Meuli, and P. Hagmann, “Predicting human resting-state functional connectivity from structural connectivity,” *Proceedings of the National Academy of Sciences*, vol. 106, no. 6, pp. 2035–2040, 2009.
- [26] H.-J. Park and K. Friston, “Structural and functional brain networks: From connections to cognition,” *Science*, vol. 342, no. 6158, Art. no. 1238411, 2013.
- [27] L. E. Suárez, R. D. Markello, R. F. Betzel, and B. Misic, “Linking structure and function in macroscale brain networks,” *Trends in Cognitive Sciences*, vol. 24, no. 4, pp. 302–315, 2020.
- [28] R. P. N. Rao and D. H. Ballard, “Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects,” *Nature Neuroscience*, vol. 2, no. 1, pp. 79–87, 1999.
- [29] K. Friston, “The free-energy principle: A unified brain theory?” *Nature Reviews Neuroscience*, vol. 11, no. 2, pp. 127–138, 2010.
- [30] T. Parr, G. Pezzulo, and K. J. Friston, *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. Cambridge, MA, USA: MIT Press, 2022.
- [31] G. Pezzulo, F. Rigoli, and K. J. Friston, “Active inference, homeostatic regulation and adaptive behavioural control,” *Progress in Neurobiology*, vol. 134, pp. 17–35, 2015.
- [32] G.-Q. Bi and M.-M. Poo, “Synaptic modifications in cultured hippocampal neurons: Dependence on spike timing, synaptic strength, and postsynaptic cell type,” *Journal of Neuroscience*, vol. 18, no. 24, pp. 10464–10472, 1998.
- [33] P. U. Diehl and M. Cook, “Unsupervised learning of digit recognition using spike-timing-dependent plasticity,” *Frontiers in Computational Neuroscience*, vol. 9, Art. no. 99, 2015.
- [34] M. Davies *et al.*, “Loihi: A neuromorphic manycore processor with on-chip learning,” *IEEE Micro*, vol. 38, no. 1, pp. 82–99, 2018.
- [35] S. B. Furber, F. Galluppi, S. Temple, and L. A. Plana, “The SpiNNaker project,” *Proceedings of the IEEE*, vol. 102, no. 5, pp. 652–665, 2014.
- [36] Anonymous Authors, “Towards ARSPI-Net: Development of an efficient hybrid deep learning framework,” in *Proc. IEEE Long Island Systems, Applications and Technology Conf. (LISAT)*, Old Westbury, NY, USA, 2023, pp. 1–8, doi: 10.1109/LISAT58403.2023.10179592.
- [37] Anonymous Authors, “Towards ARSPI-Net: Advancing EEG feature extraction with neuromorphic algorithms,” in *Proc. IEEE Long Island Systems, Applications and Technology Conf. (LISAT)*, Holtsville, NY, USA, 2024, pp. 1–8, doi: 10.1109/LISAT63094.2024.10808138.